Pressure Vessel Thickness Report

ASME BPVC Section VIII Division 1 | Internal pressure on the concave side

DESIGN EQUATION

ASME BPVC VIII Division 1, internal pressure
UG-27(c)(1),(2)
t_h = P R/(S E - 0.6P); t_l = P R/(2 S E + 0.4P)
t = max(t_h,t_l)
t_order = max(t, 1.5 mm) + CA
Pressure equations: P and S in MPa; dimensions in mm; input bar x 0.1 = MPa.

Project	QA — مثال وعاء
Prepared by	Device preview
Design reference	Internal pressure example
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01 COMPONENT AND GEOMETRY

Component	Cylindrical shell, longitudinal seam
Code rule	UG-27(c)(1) and UG-27(c)(2)
Inside radius R (corroded)	500 mm
Inside diameter D	1000 mm

02 DESIGN INPUTS

Material / product	User-supplied; not recorded
Design metal temperature	20 C
Stress source	Section II-D 1A/1B; S supplied at material/temperature
Internal design gauge pressure P	10 bar(g) = 1 MPa
Allowable stress S	138 MPa
Joint efficiency E (Table UW-12)	1
Corrosion allowance	3 mm

03 CALCULATION TRACE

t = P x R / (S x E - 0.6 x P) for the circumferential stress at a longitudinal seam, and t = P x R / (2 x S x E + 0.4 x P) for the longitudinal stress at a circumferential seam.

Circumferential stress, UG-27(c)(1)	3.639 mm
Longitudinal stress, UG-27(c)(2)	1.809 mm
Governing pressure thickness t	3.639 mm
UG-16(b) minimum after forming	1.5 mm (excluding corrosion)
Minimum thickness to order	6.639 mm (max(t, minimum formed thickness) + corrosion)

04 MAXIMUM ALLOWABLE WORKING PRESSURE

Nominal plate thickness	8 mm
MAWP for that thickness	13.7177 bar(g)

05 LIMITS AND REFERENCES

Basis: ASME BPVC Section VIII Division 1, UG-27(c) and UG-32(d)(e)(f)(g), with the UG-16(b) minimum formed thickness. S (Section II, Part D), E (Table UW-12) and the corrosion allowance are user inputs. Dimensions are stated in the corroded condition and the corrosion allowance is added to the computed thickness once. External pressure (UG-28), openings and reinforcement (UG-36/37/40), bolted flanges (Appendix 2), supports, fatigue, thermal discontinuity stress, the Division 2 alternative rules, and the thick-wall rules of Appendix 1-2 are outside this report.